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Artificial Intelligence Lab: CSE 4618 **Lab 3 - Constraint Satisfaction Problem**

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1 Introduction

In this lab, we had to create simple CSPs and solve them using the applet provided by **AIspace**.

2 Question 1 - Map Coloring

We want to color each province either Red(R),Yellow(Y),or Blue(B) in such a way that no two neighboring regions(provinces that share their borders) have the same color.

2.1 Problem Analysis

2.1.1 Variable

There were 6 variables: North Sulawesi(NS), Gorontalo(G), Central Sulawesi (CS), West Sulawesi(Ws), South Sulawesi(SS), SouthEast Sulawesi(SES).

2.1.2 Domain

Each of the variable had three values in their domain - Red(R), Blue(B) and Yellow(Y).

2.1.3 Constraints

As given, no two neighboring regions can have the same color. So all the neighbors are in "Not same" constraint.So I set the matching pairs to false like R,R,Y,Y and B,B for each neighboring region.

2.2 Solution

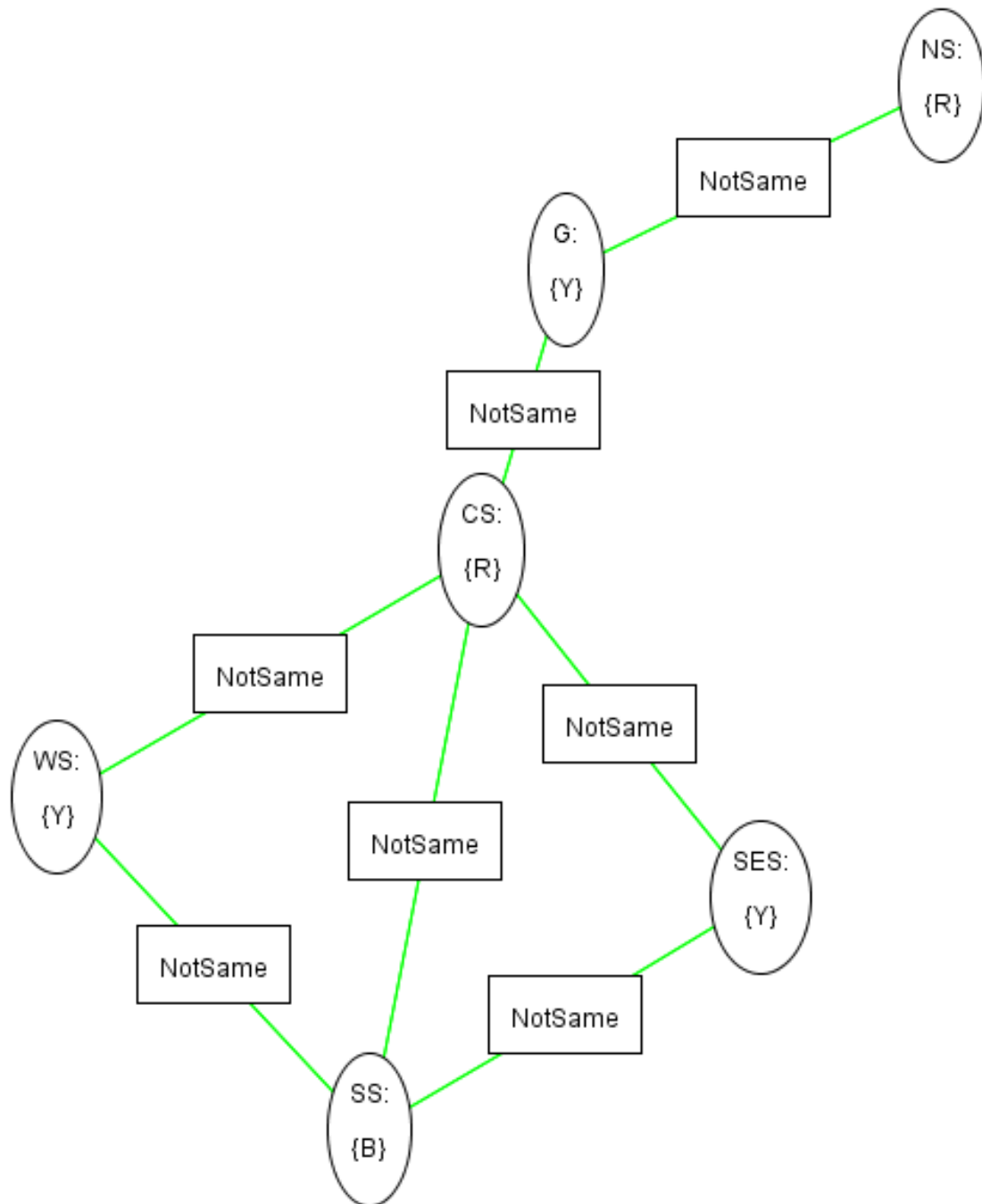


Figure 1: CSP of question 1

2.3 Findings

Solution found: $NS = R$, $G = Y$, $CS = R$, $WS = Y$, $SS = B$, $SES = Y$

2.4 Challenges

This problem was the easiest one of them, so I didn't face any challenges with this.

3 Question 2 - Eating Out

Zahid (Z), Ishrak (I), Farabi (F), and Nafisa (N) came to Chini-Come, a restaurant near their university. The restaurant serves Special Rice (S), Biryani Rice (B), Kashmiri Naan (K), and Paratha (P). The task was to create CSP based on the given constraints and come up with a solution.

3.1 Problem Analysis

3.1.1 Variable

There were four variables- F,N,Z and I

3.1.2 Domain

Each of the variable had four values in their domain- Special Rice (S), Biryani Rice (B), Kashmiri Naan (K), and Paratha (P).

3.1.3 Constraint

- As given, Z does not like Paratha. So Z can not be P. So I created a unary constraint named $Z \neq P$ and in the truth table I unchecked the value of P and set it to False.
- Ishrak and Farabi want to order different dishes. So I created a binary constraint $I \neq F$ and set all the entries of the table to false where the entry had same value for F and I.
- Farabi will either take Special Rice or Biryani Rice. So in the truth table I set only the values of B and S to true.
- Zahid will order the same dish as Ishrak. So I created a binary constraint $Z = I$ and set all the entries of the table to true where the entry had same value for Z and I.
- Also, Zahid will order a unique dish. So I created binary constraint of Z with F and N, and set all the entries of the table to false where the entry had same value for the variables.
- As Nafisa will not order K, I created a unary constraint and set the value of K for N false.

3.2 Solution

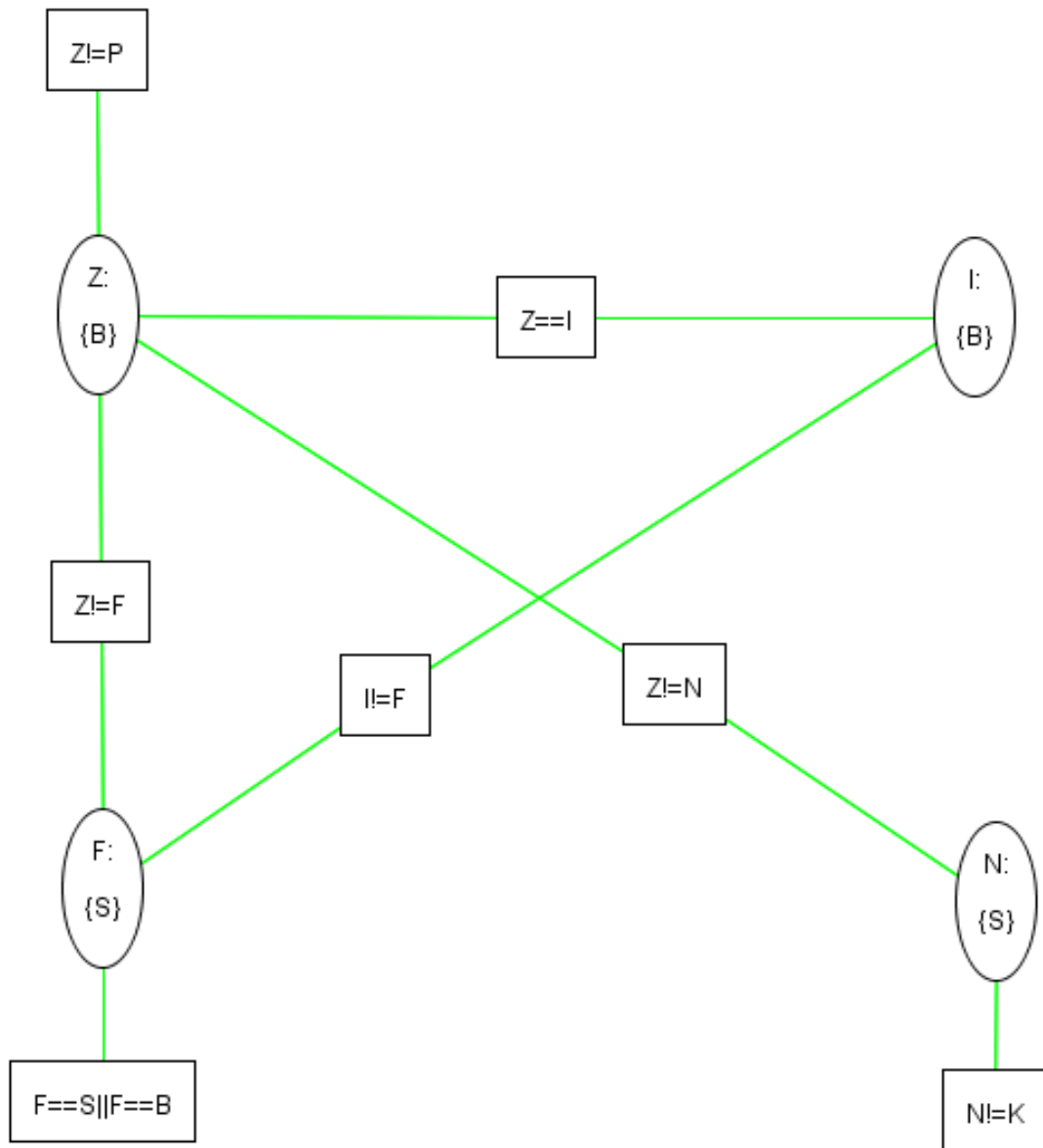


Figure 2: CSP of question 2

3.3 Findings

Solution found: $Z = B$, $F = S$, $I = B$, $N = S$

3.4 Challenges

At first I was not sure how to set the constraints for Z as he wanted unique dishes but also would order the same dish as I. So I created individual binary constraints, one with Z and I and the others for Z and F; Z and N. It worked.

4 Question 3 - Finding Houses

Four people, Ali(A), Sristy (S), Maliha (M), and Rafid (R) are looking to rent space in an apartment building. There are three floors in the building: 1, 2, and 3 (where 1 is the lowest and 3 is the highest). More than one person can live on a single floor, but each person must be assigned to some floor.

4.1 problem analysis

4.1.1 Variable

There were four variables- A,S,M and R.

4.1.2 Domain

Each of the variable had three values in their domain- 1,2 and 3 which represents floor number(Integer).

4.1.3 Constraint

- Ali and Sristy must not live on the same floor,so I took the **complement of $A==S$** using the built in constraint type Equal.
- If Ali and Maliha live on the same floor, they must both be living on floor 2. But if they live on different floors, one of them must be living on floor 3. So I set the values of **1,3, 2,2, 3,1, 2,3, and 3,2 to true**.
- Rafid must not live on the same floor as anyone else. So I created 3 binary constraints and took the **complement of $R==A$, $R==S$ and $R==M$** .
- Rafid must live on a higher floor than Maliha,so I used the **Greater Than** constraint between R and M; set $R > M$.

4.2 Solution

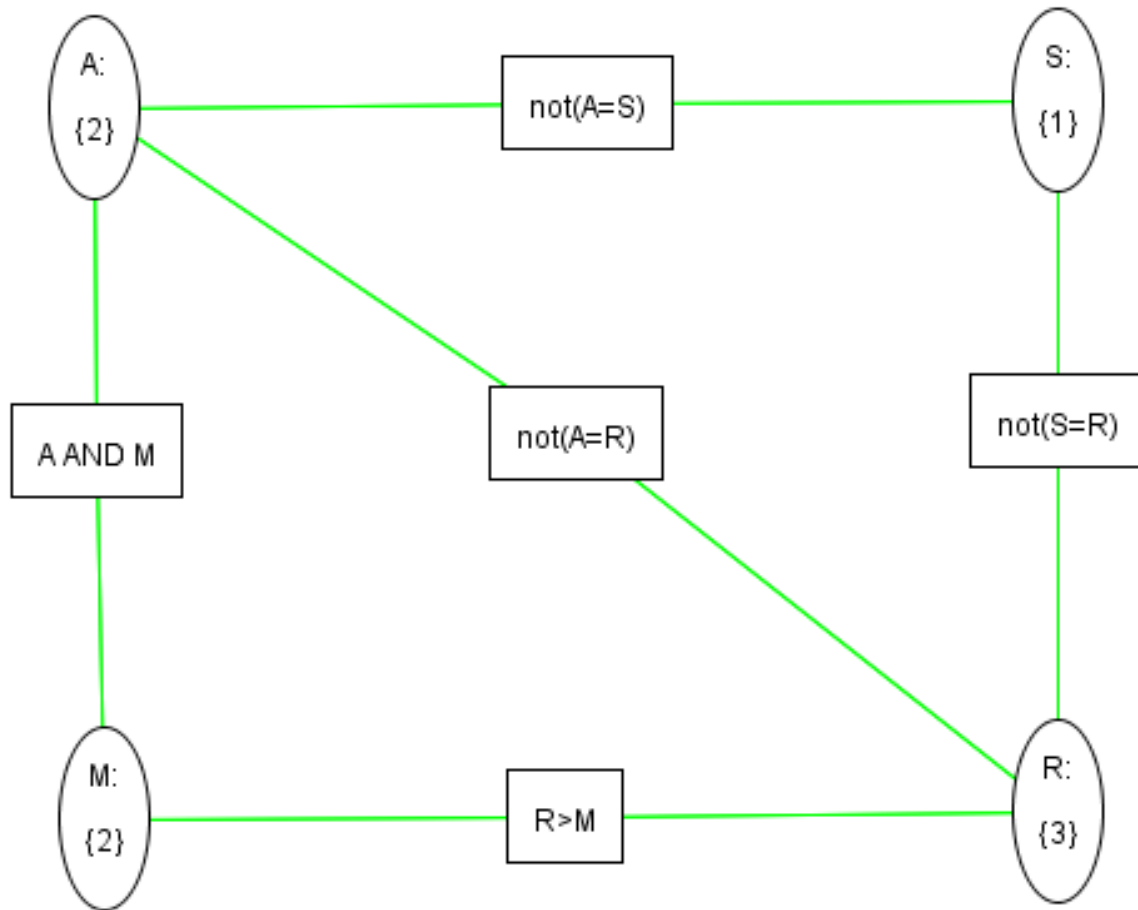


Figure 3: CSP of question 3

4.3 Findings

Solution found: $A = 2$, $S = 1$, $M = 2$, $R = 3$

4.4 Challenges

Setting up the constraint for Ali and Maliha was challenging. At first, I only set the entries true with a value of 3. But it did not meet the condition that they both must live on floor 2 if they lived on the same floor. To ensure that, later I set the value of 2,2 to true as well.

5 Question 4 - Spots

Assume that six friends Rifat (R), Atiq (A), Farhan (F), Ishmam(I), Tabassum(T), and Sabrina(S) are standing in a queue, each occupying a unique spot among the six possible spots labeled 1 to 6.

5.1 Problem Analysis

5.1.1 Variable

R,A,F,I,T,S are the variables.

5.1.2 Domain

Each variable could take values 1,2,3,4,5 and 6(Integer).

5.1.3 Constraint

- Farhan is standing in between Atiq and Ishmam. So I created two binary constraints $|F - A| = 1$ and $|F - I| = 1$. I set the values of the entries true where the difference between the pair of values were 1 such as 1,2,2,1,3,2,2,3 etc.
- Sabrina and Rifat are standing next to each other. So I created a binary constraint $|S - R| = 1$;similar to the previous 2 constraints.
- Tabassum is either at the front of the line or the back of the line. So I set created a unary constraint and set only the values of **1 and 6** to true.
- Sabrina has one person behind her. So I created a unary constraint and set only the value of **5**to true.
- Each of them were standing in unique location. So I took the complement of Equal constraint for each pair of values such as T and R, T and A, T and F, T and I, T and S and so on for other five variables.

5.2 Solution

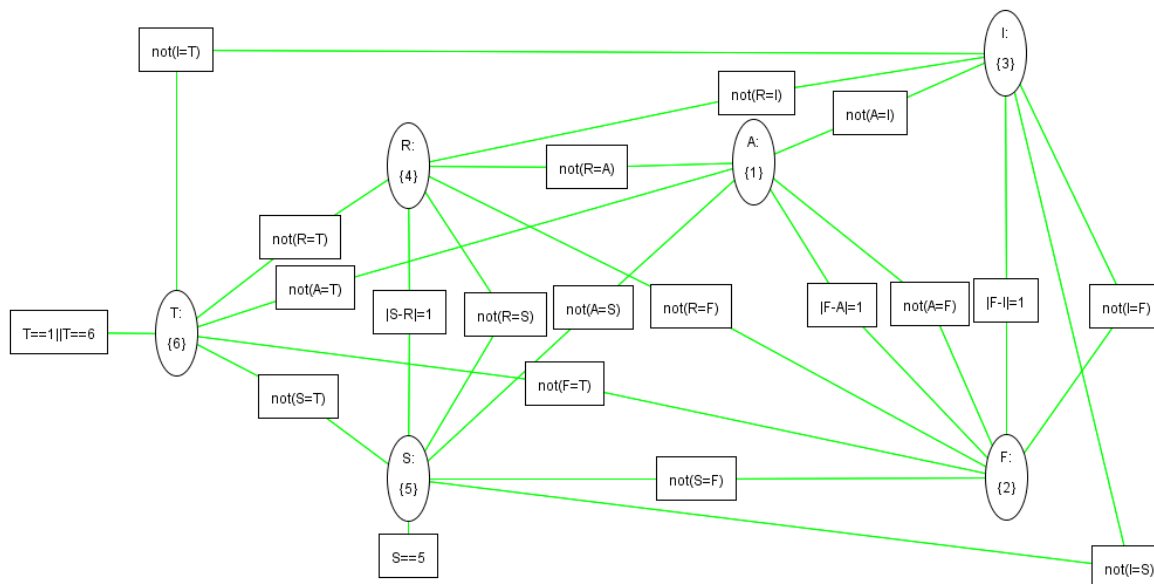


Figure 4: CSP of question 4

5.3 Findings

Solution found: $R = 4$, $A = 1$, $F = 2$, $I = 3$, $T = 6$, $S = 5$

5.4 Challenges

In this task the most difficult constraint to satisfy was that each of them was to take a unique spot. I was losing track of which ones I had already covered. I tried assigning a ternary constraint but failed to do so. That is why I created binary constraints for each pair. The resulting CSP contained too many constraints and it was massive.

6 Question 5 - Scheduling Tasks

6.1 Problem Analysis

6.1.1 Variable

The variables were G,Q,C, D, and L.

6.1.2 Domain

Each of the variables could take 8 values x8 (8 am- 9 am) , x9 (9 am- 10 am) , x10 (10 am- 11 am) , x11 (11 am- 12 pm) , y8 (8 am- 9 am) , y9 (9 am- 10 am) , y10 (10 am- 11 am) , y11 (11 am- 12 pm)

6.1.3 Constraint

- At any given time, each faculty member (G, Q, C, D, L) can do at most one task . I satisfied this constraint by creating binary constraint between each pair such as $D \neq L$, $D \neq G$, $D \neq Q$, $D \neq C$ and so on and setting false the entries of the truth table where the pair contained same values. For example x8,x8, x9,x9,x10,x10,x11,x11, y8,y8,y9,y9,y10,y10 and y11,y11.
- The AI class (C) must happen before AI lab (L). I created a binary constraint between C and L, and set the pair of values false where the values of C were greater than L.
- The contents (G) should be gathered before taking the DBMS Lab (D). I created a binary constraint between G and D, and set the pair of values false where the values of G were greater than D.
- The DBMS Lab (D) should be finished by 10 am. As DBMS lab only take 1 hour, I set the values true where the starting time was 8 and 9 for both X and Y.
- X is going to gather contents for DBMS (G) since s/he's good at browsing contents. I created a unary constraint and set the values true only for X.
- The other faculty member not conducting DBMS lab (D) should attend the lab, and hence cannot do anything else at that time. I ensured that the pair of values for X and Y where time is same are set to false. Such as x8,y8, x9,y9 etc are false.
- The person taking DBMS Lab (D) does not take AI Lab (L). I created binary constraint between D and L and the entries where both values of a pair were x or y, were set to false.
- The person taking AI Lab (L) must also take the AI class (C).I created binary constraint between C and L and the entries where both values of a pair were x or y, were set to true.
- Checking quiz scripts (Q) takes 2 consecutive hours and hence should start at or before 10 am. I created a unary constraint and set the value true where time was 8,9 and 10.

- Taking AI Lab (L) takes 2 consecutive hours and hence should start at or before 10 am. I created a unary constraint and set the value true where time was 8, 9 and 10.

6.2 Solution

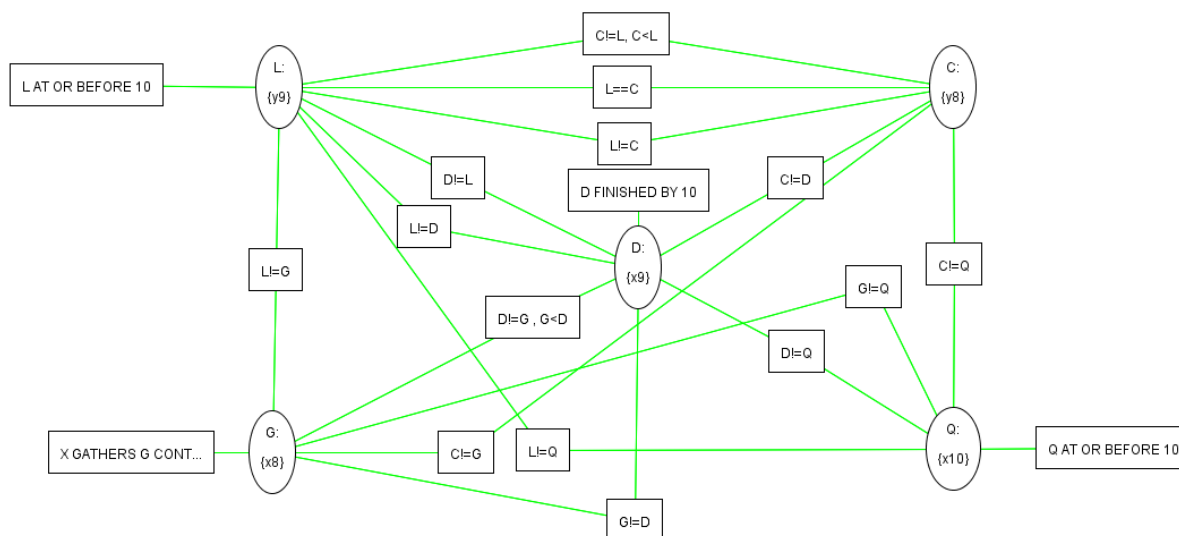


Figure 5: CSP of question 5

6.3 Findings

Solution found: $L = y9$, $D = x9$, $C = y8$, $Q = x10$, $G = x8$

6.4 Challenges

This was the hardest task for me to design the csp for. Even choosing the domain values was hard. I took some help from the instructor.

The constraints were confusing to me at first, but it became clearer as I drew the graph with pen and paper.

7 Extra Task

You are the lead event planner for a high-stakes diplomatic gathering among 8 nations: Federated States of Columbia, Zhonghua, Tsardom of Vostok, Union of Albion, Republic of Lurette, Sundhara, Kingdom of Verde, and Zayrad. Your most critical task is arranging the seating at a circular table (S1,S2,...,S8) for the 8 influential ambassadors representing each country. The table is circular, meaning seat S8 is adjacent to S1.

7.1 Problem Analysis

7.1.1 Variable

There were 8 variables : Columbia, Zhonghua,Vostok,Albion,Lurette, Sundhara,Verde, and Zayrad.

7.1.2 Domain

Each variable had values 1-8(integer) in its domain.

7.1.3 Constraint

- The ambassador from the Federated States of Columbia, as the host nation's representative, must be seated at the first seat. So I created a unary constraint and assigned $Columbia = 1$ and set all the other values to false.
- The ambassador from the Federated States of Columbia cannot be seated next to the ambassadors from the Tsardom of Vostok, Zhonghua, or Zayrad. I created binary constraints with Columbia and each of the ambassadors and set $|Columbia - Ambassador| \neq 1$.
- The ambassador from Sundhara cannot be seated next to the ambassador from Zhonghua. This is like the previous constraint. I did $|Sundhara - Zhonghua| \neq 1$.
- The ambassador from the Federated States of Columbia must sit next to the ambassador from the Union of Albion or the Republic of Lurette. So I did $|Columbia - Albion| == 1$ and $|Columbia - Lurette| == 1$.
- The ambassador from Kingdom of Verde needs to be seated next to at least one ambassador who can facilitate communication (Federated States of Columbia, Republic of Lurette, Sundhara, or Zayrad). This is like the previous constraint. So I created binary constraint between Verde and each of the facilitators and I set $|Verde - Facilitator| == 1$.
- No more than two ambassadors from the permanent members of the Global Handshake Syndicate's (GHS) Security Council (Federated States of Columbia, Zhonghua, Tsardom of Vostok, Union of Albion, Republic of Lurette) can be seated in any consecutive sequence of three seats. I couldn't do this in the csp solver. But I thought about

creating a constraint and adding three variables to it by making combinations of three nations and checking any consecutive combinations of seats to false.

7.2 Solution

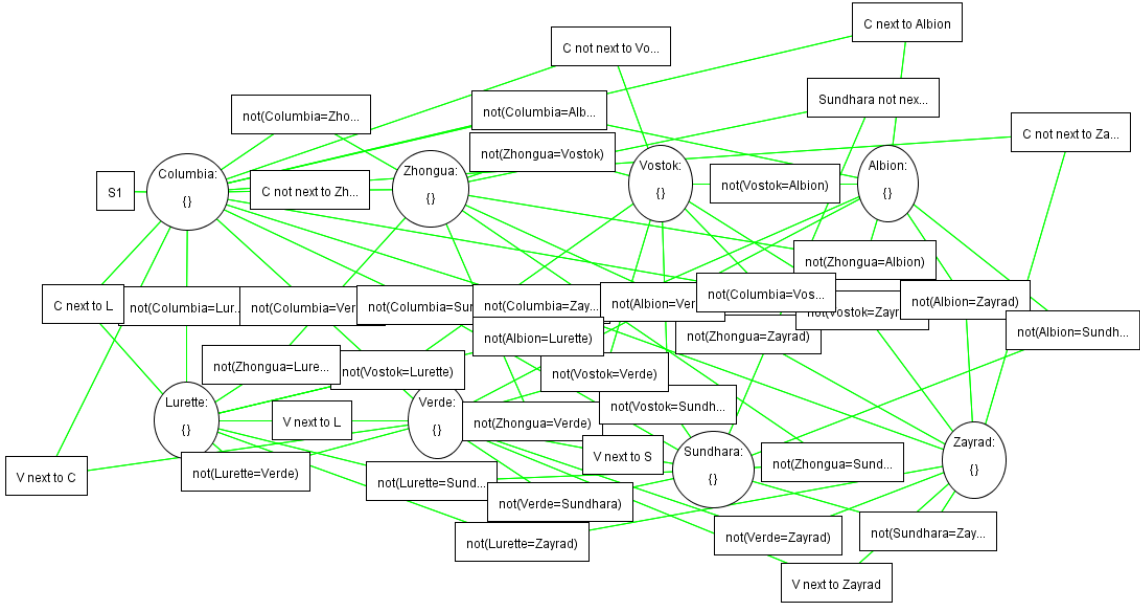


Figure 6: CSP of extra task

7.3 Findings

As I couldn't correctly implement all the constraints and maybe some other constraints were also misapplied, I didn't get any solution from the solver.

7.4 Challenges

This was the most confusing to design in the csp solver. The graph got way too big and I couldn't keep track of the constraints. I couldn't implement the last constraint.